A FINITELY LOCALLY FREE MODULE TRIVIAL ON EVERY FIBRE COMES FROM THE BASE OVER A REDUCED BASE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 9311-9324 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 9326-9331, the complete author proof (4 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The diagonal of the Picard functor for flat proper families, fibrewise triviality of a finite locally free module, and reducedness of the base; the argument is retained.

Source reference. Original lemma lemma-diagonal-picard-flat-proper: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 9199–9215. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to S$ be a flat, proper morphism of finite presentation such that $f_*\mathcal{O}_X = \mathcal{O}_S$ and this remains true after arbitrary base change. Let $\mathcal{E}$ be a finite locally free $\mathcal{O}_X$-module. Assume*

(1) *$\mathcal{E}|_{X_s}$ is isomorphic to $\mathcal{O}_{X_s}^{\oplus r_s}$ for all $s \in S$, and*
(2) *S is reduced.*

Then $\mathcal{E} = f^*\mathcal{N}$ for some finite locally free $\mathcal{O}_S$-module $\mathcal{N}$.

Proof. Namely, in this case the locally closed immersion $j : Z \to S$ of Lemma lemma-diagonal-picard-flat-proper is bijective and hence a closed immersion. But since S is reduced, j is an isomorphism. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.